

STAT293 - 11.draft

Nancy Lopez

2025-11-29

```
rm(list = ls()) # Clear all variables/objects from the environment
graphics.off()  # Close all graphics devices/plot windows
gc()            # Force garbage collection to free up memory
```

```
##          used (Mb) gc trigger (Mb) limit (Mb) max used (Mb)
## Ncells  571218 30.6   1274238 68.1         NA   715625 38.3
## Vcells 1102986  8.5    8388608 64.0        16384 2010639 15.4
```

Step 1: Packages

Step 2: Simulation Paramaters

```
rseed <- 123 #random seed
set.seed(123)

# Dimensions
K <- 20      # Number of individuals
T_len <- 30  # Total number of time points
d <- 20      # Number of variables (Dimension)

# Model Structure
p <- 1       # VAR lag order
sparsity <- 0.05 # Sparsity level (proportion of non-zero coefficients)
heterogeneity <- "medium" # Level of heterogeneity ("low", "medium", "high")

# Data Generation & Noise
sigma_ep <- 0.1 # Standard deviation of noise
phi_max <- 0.9  # Upper bound for coefficient generation
phi_min <- 0.1  # Lower bound for coefficient generation

# Evaluation Settings
n_holdout <- 5 # Number of time points withheld for forecasting (holdout)

# Code below (do not modify)
# Heterogeneity
frac_common <- switch(heterogeneity,
                      "low" = 2/3,
```

```

        "medium" = 1/2,
        "high"   = 1/3)

# total number of nonzero elements in each d × d transition matrix
n_nonzero_total <- max(1, round(sparsity * d * d))

# number of nonzero elements for common effects
n_common <- max(1, round(frac_common * n_nonzero_total))

# number of nonzero elements for unique (individual-specific) effects
n_unique <- n_nonzero_total - n_common

```

Step 3: Helper Function

Companion Matrix (F)

$$\mathbf{F} = \begin{bmatrix} \Phi_1 & \Phi_2 & \cdots & \Phi_{p-1} & \Phi_p \\ \mathbf{I}_d & \mathbf{0} & \cdots & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{I}_d & \cdots & \mathbf{0} & \mathbf{0} \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ \mathbf{0} & \mathbf{0} & \cdots & \mathbf{I}_d & \mathbf{0} \end{bmatrix}$$

```

# Generate Coefficient Matrix (Can be used for common or unique)
gen_coef_matirx <- function(d, idx, min_val = phi_min, max_val = phi_max){

  M <- matrix(0, d, d)
  n_nonzero <- length(idx)
  M[idx] <- runif(n_nonzero, min_val, max_val)
  # M[idx] <- runif(n_nonzero, min_val, max_val) * sample(c(-1,1), n_nonzero, replace = TRUE)

  return(M)
}

# Determine Stability
check_phi_stability <- function(phi_array) {
  # Require input dimensions to be [row, col, lag]
  # Ensure input is a 3D array [d, d, p].
  if (!is.array(phi_array)) {
    phi_array <- array(phi_array, dim = c(d, d, 1))
  }

  d <- dim(phi_array)[1]
  p <- dim(phi_array)[3]

  # Convert 3D Array slices into a List
  phi_list <- lapply(1:p, function(i) phi_array[, , i])

  # --- the Companion Matrix (F) ---
  # The structure is a (p*d) x (p*d) matrix:

```

```

# F = [ Phi1  Phi2 ... Phip ]
#      [ I    0    ...  0 ]
#      [ 0    I    ...  0 ]
#      ...

# 1. Construct Companion Matrix (F)
# Create Top Block: [Phi1, Phi2, ..., Phip]
top_block <- do.call(cbind, phi_list)

if (p > 1) {
  # Identity matrix on the sub-diagonal
  ident_block <- diag(1, d * (p - 1))
  # Zero column on the right
  zero_col <- matrix(0, nrow = d * (p - 1), ncol = d)
  bottom_block <- cbind(ident_block, zero_col)

  # Combine
  companion_matrix <- rbind(top_block, bottom_block)
} else {
  # VAR(1) case: Companion matrix is just Phi1 itself
  companion_matrix <- top_block
}

# 2. Calculate Eigenvalues & Determine Stability
eigen_values <- eigen(companion_matrix)$values
max_eigen <- max(Mod(eigen_values))

return(list(
  is_stable = max_eigen < 1,
  max_eigen = max_eigen
))
}

```

Step 4: Generate Individual Matrix (pd)*d

Generate Common Coefficients Matrix and Indices

```

# Position Indices
# Dimensions: [Number of Unique Elements, Subject, Lag]
# This stores the specific grid locations for each subject's unique effects
unique_indices_array <- array(0, dim = c(n_unique, K, p))

# Common Coefficients Matrix (mu)
# Dimensions: [Row, Col, Lag]
# This stores the shared coefficient matrices for the population
common_mu_array <- array(0, dim = c(d, d, p))

# Individual Matrix (B)
# Dimensions: [Row, Col, Lag, Subject]
# This will store the final stable transition matrices for every subject
individual_B <- array(0, dim = c(d, d, p, K))

```

```

for (lag in 1:p) {

  # 1. Generate Common Coefficients Matrix
  # Randomly select grid points for the common structure
  common_idx <- sample(1:(d*d), n_common)
  common_mu_array[,lag] <- gen_coef_matirx(d, idx = common_idx)

  # 2. Allocate Unique Locations
  available_pool <- setdiff(1:(d*d), common_idx)

  # Sample indices for all K subjects at once from the remaining pool
  unique_idx <- sample(available_pool, n_unique * K, replace = FALSE)

  # Reshape into a matrix where each column represents one subject
  unique_indices_array[,lag] <- matrix(unique_idx, nrow = n_unique, ncol = K)

}

```

Generation Individual Matrix and Stability Check

```

for (k in 1:K) {

  is_stable <- FALSE
  attempt <- 0
  max_attempts <- 100

  while(!is_stable) {

    attempt <- attempt + 1

    # Temporary container for this subject's p matrices
    current_subject_phi <- array(0, dim = c(d, d, p))

    for (lag in 1:p) {
      # Retrieve Common Matrix (mu)
      common_mu <- common_mu_array[,lag]

      # Looks up the specific locations pre-allocated for subject k at this lag
      unique_idx <- unique_indices_array[, k, lag]

      # Generate Unique Delta Matrix
      unique_Delta <- gen_coef_matirx(d, idx = unique_idx)

      # Phi Matrix at this lag
      current_subject_phi[,lag] <- common_mu + unique_Delta
    }

    # Stability Check
    check_result <- check_phi_stability(current_subject_phi)
  }
}

```

```

if (check_result$is_stable) {
  # Success
  is_stable <- TRUE
  individual_B[, , k] <- current_subject_phi
} else {
  # Failure: Repeat 'while' loop and regenerate unique_Delta
  # Note: 'unique_idx' and 'common_mu' remain fixed.

  # Fallback mechanism if stability is hard to achieve
  if (attempt > max_attempts) {

    # Calculate Scaling Factor
    # The current max eigenvalue is > 1 (unstable).
    # Formula: scaling_factor = current_eigen / target_radius
    target_radius <- 0.95
    scaling_factor <- check_result$max_eigen / target_radius

    # Apply Scaling
    current_subject_phi <- current_subject_phi / scaling_factor

    # Force Stability Flag
    is_stable <- TRUE
    individual_B[, , k] <- current_subject_phi

    warning(sprintf("Subject %d unstable after %d attempts. Forced scaling applied (Factor=%.2f).")
  )
  }
}
}

```

Step 5: Generate Var(p) ILD for each individual

```

gen_VAR_p <- function(phi_array, T_len, p, d, sigma_ep, burn_in = 50) {

  # Require input dimensions to be [row, col, lag]
  # Ensure input is a 3D array [d, d, p].
  if (!is.array(phi_array)) {
    phi_array <- array(phi_array, dim = c(d, d, 1))
  }
  p <- dim(phi_array)[3]

  # Total Simulation Length (Desired Length + Burn-in)
  total_T <- T_len + burn_in

  # Initialize
  X <- matrix(0, nrow = total_T, ncol = d)

  # Initialize the first 'p' steps to start the process
  X[1:p, ] <- matrix(rnorm(p * d), nrow = p, ncol = d)
}

```

```

# Generate Data Iteratively
for (t in (p + 1):total_T) {

  # Calculate the Autoregressive component: Sum(Phi_p * X_{t-p})
  ar_term <- numeric(d) # Initialize zero vector

  for (lag in 1:p) {
    # Retrieve coefficient matrix for this lag
    phi_p <- phi_array[, , lag]

    # Retrieve past data at t-lag
    past_X <- X[t - lag, ]

    # Matrix multiplication and accumulation
    ar_term <- ar_term + (phi_p %*% past_X)
  }

  # Add random error term
  X[t, ] <- ar_term + rnorm(d, 0, sigma_ep)
}

# Remove Burn-in Period
# Return only the valid data sequence of length T_len
return(X[(burn_in + 1):total_T, ])
}

# Generate Data for All Subjects
# data_list[[k]] contains the observed data matrix for subject k
# individual_B is 4D array [d, d, p, K]

data_list <- lapply(1:K, function(k) {

  # Get coefficients for subject k
  # Dimensions: [d, d, p]
  subject_phi <- array(individual_B[, , , k], dim = c(d, d, p))

  # Generate time series data
  # Returns a matrix of size [T_len, d]
  gen_VAR_p(subject_phi, T_len, p, d, sigma_ep)
})

```

Step 6: Split Training & Test Datasets

```

# Define the split index (The end of the training period)
split_idx <- T_len - n_holdout

# Create Training Data
train_data <- lapply(data_list, function(X) { X[1:split_idx, ] })

# Create Test Data (With Lookback)

```

```

# Include the last 'p' rows of the training set.
# The "initial conditions (lags)" for the first prediction in the test set.
test_data <- lapply(data_list, function(X) { X[(split_idx + 1 - p):T_len, ] })

# --- Note for Evaluation ---
# 'test_data' now has (n_holdout + p) rows. The first 'p' rows are history.
# Evaluate metrics only on the last 'n_holdout' rows (the actual test set).

```

Step 7: (a) Individual LASSO for VAR(p)

Functions

```

# Construct the VAR(p) Design Matrix
# -----
# Inputs:
#   X: The raw time series matrix [T x d]
#   p: Number of lags
# -----
gen_design_mat <- function(X, p = 1) {
  t <- nrow(X) # Total time points (T)

  # Create Target Matrix Y (Current time t)
  Y_mat <- X[(p + 1):t, ]

  # Create Predictor Matrix X_lags (Past times)
  # Stack [ X_{t-1} | X_{t-2} | ... | X_{t-p} ] horizontally.
  lag_list <- vector("list", p)

  for (lag in 1:p) {
    # If Y ranges from [p+1] to [t],
    # then X_{t-lag} ranges from [p+1-lag] to [t-lag].
    lag_list[[lag]] <- X[(p + 1 - lag):(t - lag), ]
  }

  X_lags <- do.call(cbind, lag_list)

  Z_mat <- cbind(Y_mat, X_lags)

  return(Z_mat)
}

# Extract Estimated Coefficients from LASSO Results
# -----
# Output Dimension: [d] x [d * p]
# Structure: [ Phi_1 | Phi_2 | ... | Phi_p ] (Horizontally stacked)
# -----
extract_B_hat <- function(fit_list, d, p) {

```

```

# Initialize an empty matrix to hold coefficients
# Rows = Equations (Variables)
# Cols = All predictors across all lags
B_hat <- matrix(0, nrow = d, ncol = d * p)

# Iterate through each variable j (each equation)
for (j in 1:d) {

  model <- fit_list[[j]]

  # Extract coefficients at the optimal lambda "lambda.min"
  raw_coefs <- coef(model, s = "lambda.min")

  # Remove Intercept
  beta_vec <- as.vector(raw_coefs)[-1]

  # Safety Check: Ensure the length matches our expectation
  if (length(beta_vec) == d * p) {
    B_hat[j, ] <- beta_vec
  } else {
    warning(paste("Coefficient length mismatch for variable", j))
  }
}

return(B_hat)
}

# Reshape Flat Matrix to 3D Array
# True B Array [d, d, p].

reshape_to_array <- function(B_flat, d, p) {
  B_array <- array(0, dim = c(d, d, p))

  for (lag in 1:p) {
    # Calculate column indices for this specific lag in the flat matrix
    col_idx <- ((lag - 1) * d + 1) : (lag * d)

    # Slice the columns and store them in the 3rd dimension of the array
    B_array[, , lag] <- B_flat[, col_idx]
  }

  return(B_array)
}

# Fit Lasso for Individual VAR(p)
fit_lasso_var_p <- function(X, p = 1, lambda_ = NULL) {

  d <- ncol(X)

  Z_mat <- gen_design_mat(X, p)

```



```

# The first d columns are Y_t (Current)
Y_mat <- Z_mat[, 1:d]

# The remaining columns are the stacked lags (Past)
X_lags <- Z_mat[, -c(1:d)]

# Fit LASSO for each variable j
fit_list <- vector("list", d)

for (j in 1:d) {
  fit_list[[j]] <- cv.glmnet(
    x = X_lags,
    y = Y_mat[, j],
    alpha = 1, # alpha=1 specifies the LASSO penalty (L1 norm)
    lambda = lambda_, # If NULL, glmnet selects its own sequence
    nfolds = 5
  )
}

est_B_list <- extract_B_hat(fit_list, d, p)
est_B_array <- reshape_to_array(est_B_list, d, p)

return(est_B_array)
}

```

Fit Lasso VAR(1) and VAR(p) for each subject

```

# fit LASSO for each person in train_data

# VAR(p)
if (p != 1){
  est_lasso_B_p <- lapply(train_data, function(data) {
    fit_lasso_var_p(data, p)
  })

  est_lasso_B_p <- array(unlist(est_lasso_B_p), dim = c(d, d, p, K))
}

# VAR(1)
est_lasso_B <- lapply(train_data, function(data) {
  fit_lasso_var_p(data, p = 1)
})

est_lasso_B <- array(unlist(est_lasso_B), dim = c(d, d, K))

```

Step 8: (b) Multi-VAR

```

multivar_B <- function(est_multivar_B, d, K){
  est_multivar_B <- array(unlist(est_multivar_B), dim = c(d, d, K))
}

```

```

    return(est_multivar_B)
}

```

LASSO

```

lambda_standard <- 10^seq(log10(0.01), log10(5), length.out = 30)

multivar_lasso_model <- constructModel(
  data = train_data,
  lag = 1,
  #lambda1 = lambda_standard,
  #lambda2 = lambda_standard,
  standardize = TRUE,
  canonical = FALSE,
  lassotype = "standard", # standard multi-VAR (not adaptive)
  cv = "blocked",
  nfold = 10
)

multivar_lasso_res <- cv.multivar(multivar_lasso_model)

```

```
## |
```

```
est_multivar_lasso_B <- multivar_B(multivar_lasso_res$mats$total, d, K)
```

Adaptive

```

lambda_adaptive <- 10^seq(log10(0.01), log10(5), length.out = 30)

multivar_adapt_model <- constructModel(
  data = train_data,
  lag = 1,
  #lambda1 = lambda_adaptive,
  #lambda2 = lambda_adaptive,
  standardize = TRUE,
  #weightest = "ridge",
  canonical = FALSE,
  lassotype = "adaptive",
  cv = "blocked",
  nfold = 10
)

multivar_adapt_res <- cv.multivar(multivar_adapt_model)

```

```
## |
```

```
est_multivar_adapt_B <- multivar_B(multivar_adapt_res$mats$total, d, K)
```

Step 9: Evaluation

Error

```
B_true <- individual_B[ , , 1, ]

diff_func <- function(est_B, B_true){
  sum((est_B - B_true)^2)
}

diffB_lasso <- diff_func(est_lasso_B, B_true)
diffB_multivar_lasso <- diff_func(est_multivar_lasso_B, B_true)
diffB_multivar_adapt <- diff_func(est_multivar_adapt_B, B_true)

diff_table <- data.frame(
  Method = c("LASSO (Lag 1)", "Multi-VAR (Standard)", "Multi-VAR (Adaptive)"),
  Error_Metric = c(diffB_lasso, diffB_multivar_lasso, diffB_multivar_adapt),
  stringsAsFactors = FALSE
)

if (p != 1) {
  diffB_lasso_p <- diff_func(est_lasso_B_p, individual_B)

  extra_row <- data.frame(
    Method = paste0("LASSO (Lag ", p, ")"),
    Error_Metric = diffB_lasso_p,
    stringsAsFactors = FALSE
  )

  diff_table <- rbind(diff_table, extra_row)
}

diff_table

##           Method Error_Metric
## 1      LASSO (Lag 1)    72.79369
## 2 Multi-VAR (Standard)    41.06679
## 3 Multi-VAR (Adaptive)    58.80737

kable(diff_table, digits = 3,
      caption = "Comparison of Estimation Error")
```

Table 1: Comparison of Estimation Error

Method	Error_Metric
LASSO (Lag 1)	72.794

Method	Error_Metric
Multi-VAR (Standard)	41.067
Multi-VAR (Adaptive)	58.807

Sensitivity and Specificity

```

sens_spec_func <- function(B_true, B_est, thr = 1e-4) {
  true_nonzero <- abs(B_true) != 0
  est_nonzero <- abs(B_est) > thr

  TP <- sum(est_nonzero & true_nonzero)
  FN <- sum(!est_nonzero & true_nonzero)
  TN <- sum(!est_nonzero & !true_nonzero)
  FP <- sum(est_nonzero & !true_nonzero)

  if ((TP + FN) > 0) {
    sensitivity <- TP / (TP + FN)
  } else {
    sensitivity <- NA
  }

  if ((TN + FP) > 0) {
    specificity <- TN / (TN + FP)
  } else {
    specificity <- NA
  }

  return(list(
    sensitivity = sensitivity,
    specificity = specificity,

    TP = TP, FP = FP, TN = TN, FN = FN
  ))
}

# Calculate Mean Metrics Across All Subjects
# -----
# Inputs:
#   B_true : 3/4 Array [d, d, (p), K] containing true values
#   B_est  : 3/4D Array [d, d, (p), K] containing estimates
#   thr     : Threshold (default 1e-4)
#   output_type
# -----
calc_metrics <- function(B_true, B_est, thr = 1e-4,
                        output_type = c("mean", "subject", "both")) {

  # output_type = "mean" (default)
  output_type <- match.arg(output_type)

```

```

# Check dimensions
if (!all(dim(B_true) == dim(B_est))) {
  stop("Error: Dimensions of True and Estimated arrays do not match!")
}

dims <- dim(B_true)
ndim <- length(dims)

if (ndim == 3) {
  # CASE A: Input is 3D [d, d, K] (Implies VAR(1), p=1)
  d <- dims[1]
  K <- dims[3]

  # "Reshape" to 4D by adding a singleton dimension for p=1
  dim(B_true) <- c(d, d, 1, K)
  dim(B_est) <- c(d, d, 1, K)

} else if (ndim == 4) {
  K <- dims[4]

} else {
  stop("Error: Input arrays must be 3D [d,d,K] or 4D [d,d,p,K]")
}

sens_vec <- numeric(K)
spec_vec <- numeric(K)

# Loop through each subject
for (k in 1:K) {
  # Slice the 4D array to get the 3D block [d, d, p] for subject k
  # This block contains all lags for this person
  B_true_k <- B_true[, , , k]
  B_est_k <- B_est[, , , k]

  # Calculate metrics for this person
  res <- sens_spec_func(B_true_k, B_est_k, thr = thr)

  sens_vec[k] <- res$sensitivity
  spec_vec[k] <- res$specificity
}

# Return the Mean across all subjects
if (output_type == "mean") {
  return(list(
    mean_sens = mean(sens_vec),
    mean_spec = mean(spec_vec)
  ))
} else if (output_type == "subject") {
  return(list(
    sensitivity = sens_vec,

```

```

        specificity = spec_vec
    ))

    } else { # "both"

        return(list(
            summary = list(mean_sens = mean(sens_vec), mean_spec = mean(spec_vec)),
            raw_data = list(sens_vec = sens_vec, spec_vec = spec_vec)
        ))
    }
}

```

```

# LASSO
metrics_lasso <- calc_metrics(
  B_true,
  B_est = est_lasso_B,
  output_type = "mean"
)

# multivar lasso
metrics_multivar_lasso <- calc_metrics(
  B_true,
  B_est = est_multivar_lasso_B,
  output_type = "mean"
)

# multivar adaptive
metrics_multivar_adapt <- calc_metrics(
  B_true,
  B_est = est_multivar_adapt_B,
  output_type = "mean"
)

# results
metrics_var1 <- data.frame(
  Method= c("LASSO (Lag 1)", "Multi-VAR (Standard)", "Multi-VAR (Adaptive)"),
  Sensitivity = c(metrics_lasso$mean_sens,
                  metrics_multivar_lasso$mean_sens,
                  metrics_multivar_adapt$mean_sens),
  Specificity = c(metrics_lasso$mean_spec,
                  metrics_multivar_lasso$mean_spec,
                  metrics_multivar_adapt$mean_spec)
)

ss_dat <- metrics_var1

if (p != 1) {
  metrics_lasso_p <- calc_metrics(
    B_true = individual_B,
    B_est = est_lasso_B_p,
    output_type = "mean"
  )
}

```

```

metrics_varp <- data.frame(
  Method = "LASSO (Lag p)",
  Sensitivity = metrics_lasso_p$mean_sens,
  Specificity = metrics_lasso_p$mean_spec
)

ss_dat <- rbind(ss_dat, metrics_varp)
}

ss_dat

```

```

##           Method Sensitivity Specificity
## 1      LASSO (Lag 1)      0.645    0.8677632
## 2 Multi-VAR (Standard)      0.850    0.7948684
## 3 Multi-VAR (Adaptive)      0.460    0.9853947

```

```

kable(ss_dat, digits = 3,
      caption = "Comparison of Sensitivity and Specificity")

```

Table 2: Comparison of Sensitivity and Specificity

Method	Sensitivity	Specificity
LASSO (Lag 1)	0.645	0.868
Multi-VAR (Standard)	0.850	0.795
Multi-VAR (Adaptive)	0.460	0.985

Sensitivity and Specificity (per subject)

```

# LASSO
Smetrics_lasso <- calc_metrics(
  B_true,
  B_est = est_lasso_B,
  output_type = "subject"
)

# multivar lasso
Smetrics_multivar_lasso <- calc_metrics(
  B_true,
  B_est = est_multivar_lasso_B,
  output_type = "subject"
)

# multivar adaptive
Smetrics_multivar_adapt <- calc_metrics(
  B_true,
  B_est = est_multivar_adapt_B,
  output_type = "subject"
)

```

```

# results
ss_dat_subject <- data.frame(
  Method = rep(c("LASSO (Lag 1)", "Multi-VAR (Standard)", "Multi-VAR (Adaptive)"), each = K),
  Sensitivity = c(Smetrics_lasso$sensitivity,
                  Smetrics_multivar_lasso$sensitivity,
                  Smetrics_multivar_adapt$sensitivity),
  Specificity = c(Smetrics_lasso$specificity,
                  Smetrics_multivar_lasso$specificity,
                  Smetrics_multivar_adapt$specificity)
)

if (p != 1) {
  Smetrics_lasso_p <- calc_metrics(
    B_true = individual_B,
    B_est = est_lasso_B_p,
    output_type = "subject"
  )

  Smetrics_varp <- data.frame(
    Method = "LASSO (Lag p)",
    Sensitivity = Smetrics_lasso_p$sensitivity,
    Specificity = Smetrics_lasso_p$specificity
  )

  ss_dat_subject <- rbind(ss_dat_subject, Smetrics_varp)
}

```

```

desired_order <- c("LASSO (Lag 1)",
                  "LASSO (Lag p)",
                  "Multi-VAR (Standard)",
                  "Multi-VAR (Adaptive)")

summary_table <- ss_dat_subject %>%
  group_by(Method) %>%
  summarise(
    # --- Sensitivity ---
    Sens_Mean = mean(Sensitivity, na.rm = TRUE),
    Sens_SD   = sd(Sensitivity, na.rm = TRUE),

    # --- Specificity ---
    Spec_Mean = mean(Specificity, na.rm = TRUE),
    Spec_SD   = sd(Specificity, na.rm = TRUE)
  ) %>%

  mutate(
    `Sensitivity Mean (SD)` = sprintf("%.3f (%.3f)", Sens_Mean, Sens_SD),
    `Specificity Mean (SD)` = sprintf("%.3f (%.3f)", Spec_Mean, Spec_SD),
    Method = factor(Method, levels = desired_order)
  ) %>%

  arrange(Method) %>%

  select(Method, `Sensitivity Mean (SD)`, `Specificity Mean (SD)`)

```



```
summary_table
```

```
## # A tibble: 3 x 3
##   Method          'Sensitivity Mean (SD)' 'Specificity Mean (SD)'
##   <fct>          <chr>                  <chr>
## 1 LASSO (Lag 1)    0.645 (0.131)          0.868 (0.045)
## 2 Multi-VAR (Standard) 0.850 (0.069)          0.795 (0.011)
## 3 Multi-VAR (Adaptive) 0.460 (0.137)          0.985 (0.021)
```

```
kable(summary_table, caption = "Model Performance: Mean and Standard Deviation across Subjects")
```

Table 3: Model Performance: Mean and Standard Deviation across Subjects

Method	Sensitivity Mean (SD)	Specificity Mean (SD)
LASSO (Lag 1)	0.645 (0.131)	0.868 (0.045)
Multi-VAR (Standard)	0.850 (0.069)	0.795 (0.011)
Multi-VAR (Adaptive)	0.460 (0.137)	0.985 (0.021)

RMSFE (Root Mean Square Forecast Error)

```
compute_rmsfe_p <- function(B_est, test_data_matrix) {

  # B_est: Coefficient array for a single subject.
  #       Expected dimensions: [d, d, p] or [d, d] (if p=1)
  # test_data_matrix: The test data including the initial lags.
  #                   Structure: Rows 1 to p are the "Lags" (from Training end).
  #                   Rows (p+1) to End are the "Targets" (Test data).

  # Ensure B_est is a 3D array [d, d, p]
  # If B_est is a 2D matrix (implying p=1), convert it to 3D array [d, d, 1]
  if (is.matrix(B_est) || length(dim(B_est)) == 2) {
    d_dim <- nrow(B_est)
    B_est <- array(B_est, dim = c(d_dim, d_dim, 1))
  }

  d <- dim(B_est)[1]      # Number of variables
  p <- dim(B_est)[3]      # Lag order
  n_total <- nrow(test_data_matrix) # Total rows in test set (including lags)

  # Define Actuals (Targets)
  # The actual values predicted start from row (p + 1)
  actuals <- test_data_matrix[(p + 1):n_total, ]

  # Initialize Prediction Container
  # Size matches the number of actual target observations
  preds <- matrix(0, nrow = nrow(actuals), ncol = ncol(actuals))

  # Rolling One-Step Ahead Forecast
```

```

# Predict  $X_t$  based on  $X_{t-1}$ , ...,  $X_{t-p}$ 
# Loop  $t$  runs through the indices of the original test_data_matrix
for (t in (p + 1):n_total) {

  # Initialize prediction vector for time  $t$ 
  pred_t <- numeric(d)

  # Accumulate contributions from each lag:  $\text{Sum}(A_{\text{lag}} * X_{t-\text{lag}})$ 
  for (lag in 1:p) {
    A_l <- B_est[, , lag] # Coefficient matrix for lag 'l'
    x_lag <- test_data_matrix[t - lag, ] # Data vector at time  $t-l$ 

    pred_t <- pred_t + (A_l %*% x_lag)
  }

  # Store prediction
  # Note: preds index starts at 1, corresponding to  $t-(p)$ 
  preds[t - p, ] <- pred_t
}

# Calculate Root Mean Squared Forecast Error
return(sqrt(mean((preds - actuals)^2)))
}

get_rmsfe_p <- function(B_est_obj, test_data_list, K) {

  # Initialize vector to store RMSFE for each subject
  rmsfe_vec <- numeric(K)

  for (k in 1:K) {

    # --- Logic to extract coefficients for Subject  $k$  ---

    # Case 1: Input is a 4D Array  $[d, d, p, K]$  (e.g., Multi-VAR output)
    if (is.array(B_est_obj) && length(dim(B_est_obj)) == 4) {
      B_k <- B_est_obj[, , , k]

      # Case 2: Input is a 3D Array  $[d, d, K]$  (e.g., Multi-VAR output when  $p=1$ )
    } else if (is.array(B_est_obj) && length(dim(B_est_obj)) == 3) {
      # Extract  $k$ -th slice and ensure it keeps array structure for compute_rmsfe_p
      B_temp <- B_est_obj[, , k]
      # Convert to  $[d, d, 1]$  so dimension checks pass
      B_k <- array(B_temp, dim = c(nrow(B_temp), ncol(B_temp), 1))

      # Case 3: Input is a List (e.g., LASSO output)
    } else if (is.list(B_est_obj)) {
      B_k <- B_est_obj[[k]]

    } else {
      # Case 4: Pooled/Shared model (Same matrix for everyone)
      B_k <- B_est_obj
    }
  }
}

```

```

    }

    # --- Compute RMSFE ---
    # Pass the extracted coefficients and the test data (which contains lags)
    rmsfe_vec[k] <- compute_rmsfe_p(B_k, test_data_list[[k]])
  }

  # Return the average RMSFE across all subjects
  return(rmsfe_vec)
}

# LASSO
rmsfe_lasso <- get_rmsfe_p(est_lasso_B, test_data, K)

# Multi-VAR (Standard)
rmsfe_multivar_lasso <- get_rmsfe_p(est_multivar_lasso_B, test_data, K)

# Multi-VAR (Adaptive)
rmsfe_multivar_adapt <- get_rmsfe_p(est_multivar_adapt_B, test_data, K)

rmsfe_dat <- data.frame(
  Method = rep(c("LASSO (Lag 1)", "Multi-VAR (Standard)", "Multi-VAR (Adaptive)"), each = K),
  RMSFE = c(rmsfe_lasso, rmsfe_multivar_lasso, rmsfe_multivar_adapt))

if (p != 1) {
  rmsfe_lasso_p <- get_rmsfe_p(est_lasso_B_p, test_data, K)

  rmsfe_dat_varp <- data.frame(
    Method = "LASSO (Lag p)",
    RMSFE = rmsfe_lasso_p
  )

  rmsfe_dat <- rbind(rmsfe_dat, rmsfe_dat_varp)
}

# results
rmsfe_table <- rmsfe_dat %>%
  group_by(Method) %>%
  summarise(
    Mean = mean(RMSFE, na.rm = TRUE),
    SD = sd(RMSFE, na.rm = TRUE),
  ) %>%

  mutate(
    `RMSFE Mean (SD)` = sprintf("%.3f (%.3f)", Mean, SD),
  ) %>%

  arrange(Method) %>%

  select(Method, `RMSFE Mean (SD)`)

```

```
rmsfe_table
```

```
## # A tibble: 3 x 2
##   Method      'RMSFE Mean (SD)'
##   <chr>      <chr>
## 1 LASSO (Lag 1)      0.112 (0.011)
## 2 Multi-VAR (Adaptive) 0.112 (0.015)
## 3 Multi-VAR (Standard) 0.106 (0.009)
```

```
kable(rmsfe_table, caption = "Model Performance: Mean and Standard Deviation across Subjects")
```

Table 4: Model Performance: Mean and Standard Deviation across Subjects

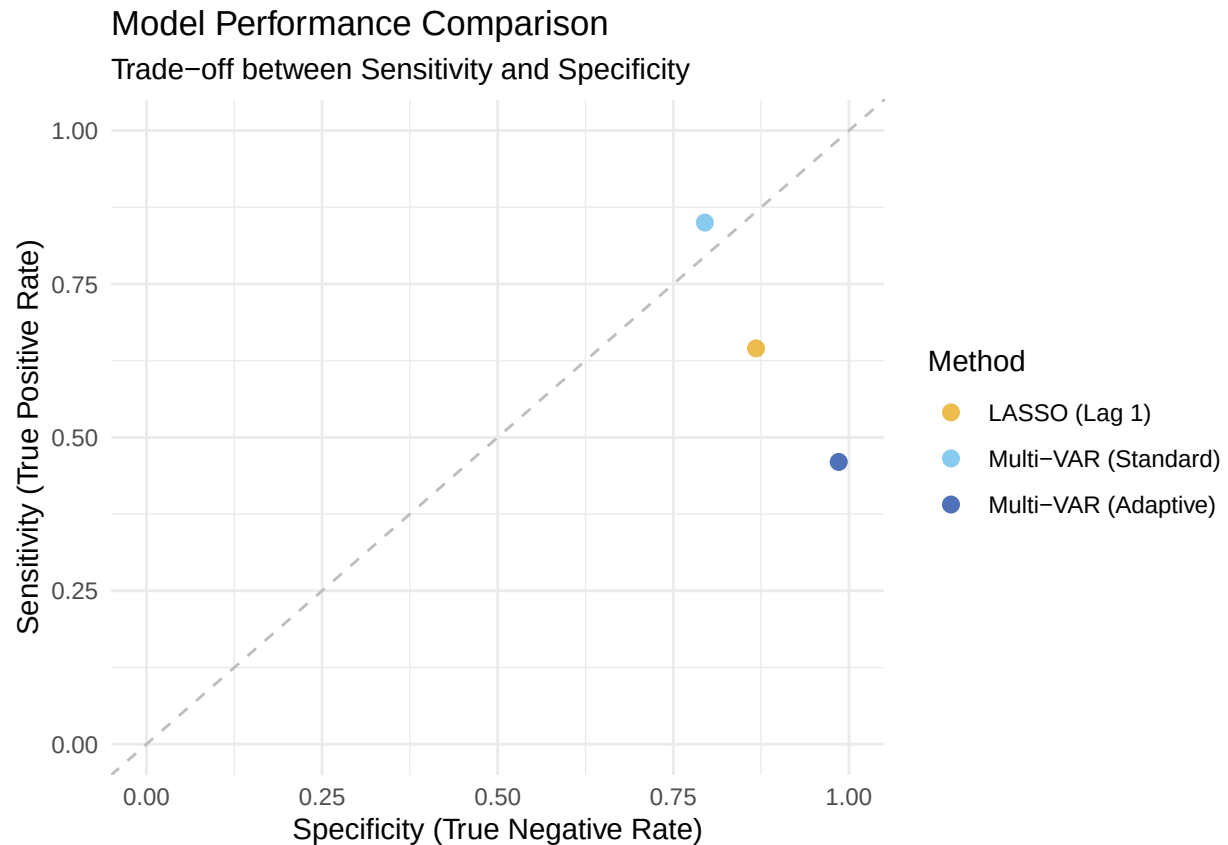
Method	RMSFE Mean (SD)
LASSO (Lag 1)	0.112 (0.011)
Multi-VAR (Adaptive)	0.112 (0.015)
Multi-VAR (Standard)	0.106 (0.009)

Step 10: Visualization

Sensitivity and Specificity (mean)

```
ss_dat$Method <- factor(ss_dat$Method, levels = desired_order)
color_lab <- c("LASSO (Lag 1)" = "#E69F00",
              "LASSO (Lag p)" = "#CC3300",
              "Multi-VAR (Standard)" = "#56B4E9",
              "Multi-VAR (Adaptive)" = "#003399")

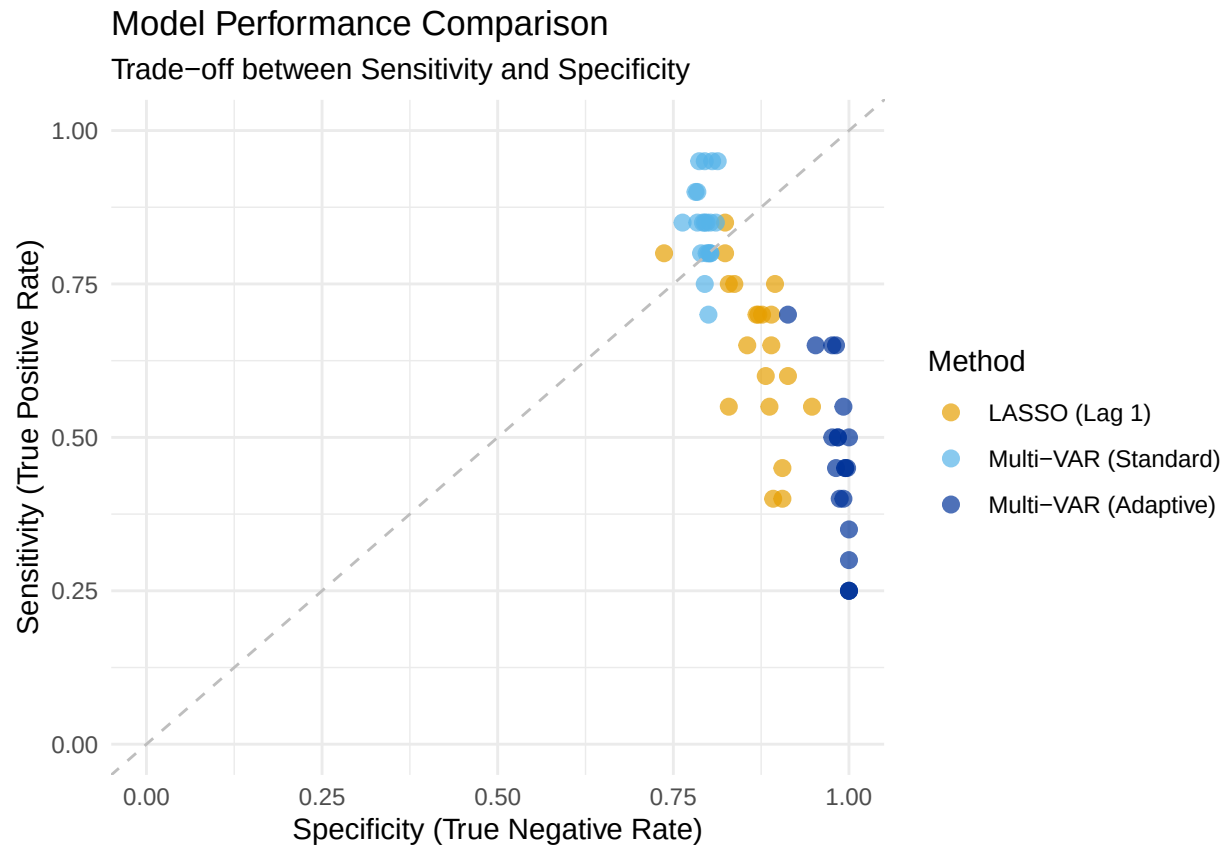
ggplot(ss_dat, aes(x = Specificity, y = Sensitivity, color = Method)) +
  geom_point(size = 2.5, alpha = 0.7) +
  xlim(0, 1) +
  ylim(0, 1) +
  geom_abline(intercept = 0, slope = 1, linetype = "dashed", color = "gray") +
  labs(
    title = "Model Performance Comparison",
    subtitle = "Trade-off between Sensitivity and Specificity",
    x = "Specificity (True Negative Rate)",
    y = "Sensitivity (True Positive Rate)"
  ) +
  theme_minimal() +
  scale_color_manual(values = color_lab)
```



Sensitivity and Specificity (per subject)

```
ss_dat_subject$Method <- factor(ss_dat_subject$Method, levels = desired_order)

# scatter plot
ggplot(ss_dat_subject, aes(x = Specificity, y = Sensitivity, color = Method)) +
  geom_point(size = 2.5, alpha = 0.7) +
  xlim(0, 1) +
  ylim(0, 1) +
  geom_abline(intercept = 0, slope = 1, linetype = "dashed", color = "gray") +
  labs(
    title = "Model Performance Comparison",
    subtitle = "Trade-off between Sensitivity and Specificity",
    x = "Specificity (True Negative Rate)",
    y = "Sensitivity (True Positive Rate)"
  ) +
  theme_minimal() +
  scale_color_manual(values = color_lab)
```



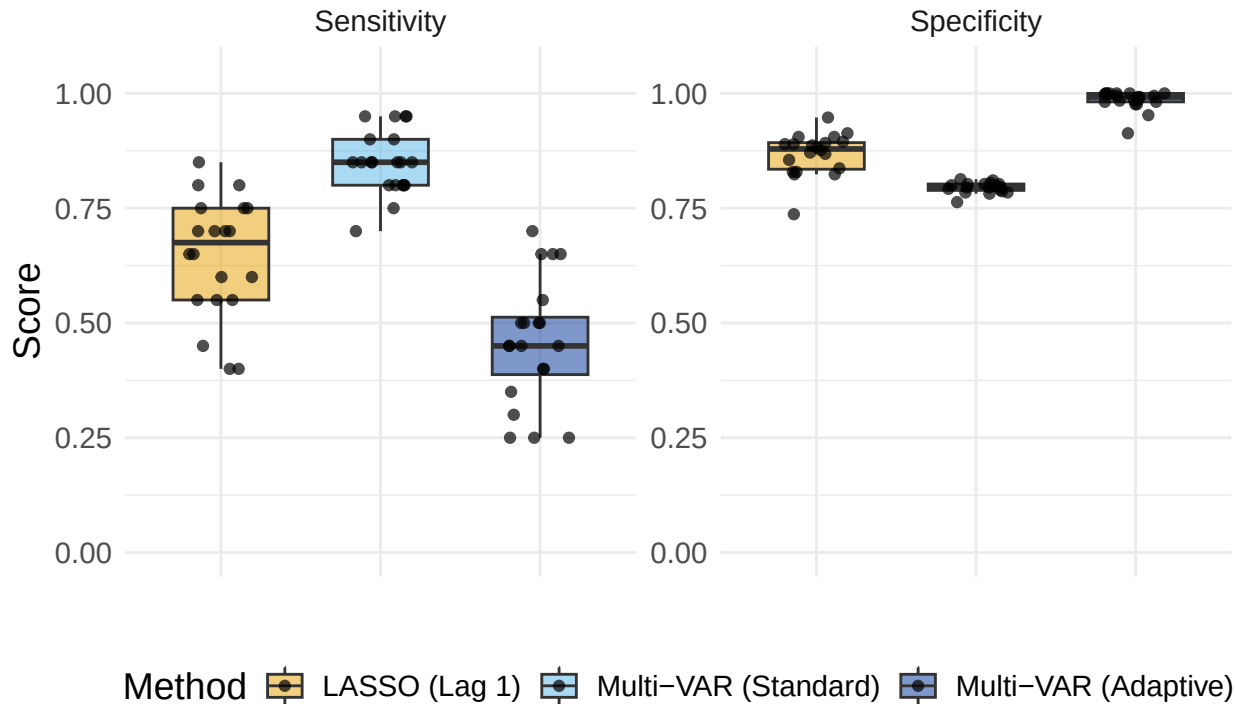
```
# boxplot
ss_dat_subject_long <- melt(ss_dat_subject,
  id.vars = c("Method"),
  measure.vars = c("Sensitivity", "Specificity"),
  variable.name = "Metric",
  value.name = "Score")
ss_dat_subject_long$Method <- factor(ss_dat_subject_long$Method, levels = desired_order)

ggplot(ss_dat_subject_long, aes(x = Method, y = Score, fill = Method)) +
  geom_boxplot(alpha = 0.5, outlier.shape = NA, width = 0.6) +
  geom_jitter(width = 0.2, height = 0, size = 1.5, alpha = 0.7) +
  facet_wrap(~Metric, scales = "free_y") +
  scale_fill_manual(values = color_lab) +
  labs(
    title = "Model Performance Distribution (Per Subject)",
    subtitle = "Boxplots show variability across 10 subjects",
    y = "Score",
    x = ""
  ) +
  ylim(0, 1.05) +
  theme_minimal() +
  theme(
    axis.text.x = element_blank(),
    axis.ticks.x = element_blank(),
    legend.position = "bottom",
    text = element_text(size = 14)
  )
```

)

Model Performance Distribution (Per Subject)

Boxplots show variability across 10 subjects



RMSFE (per subject)

```
# boxplot
rmsfe_dat$Method <- factor(rmsfe_dat$Method, levels = desired_order)

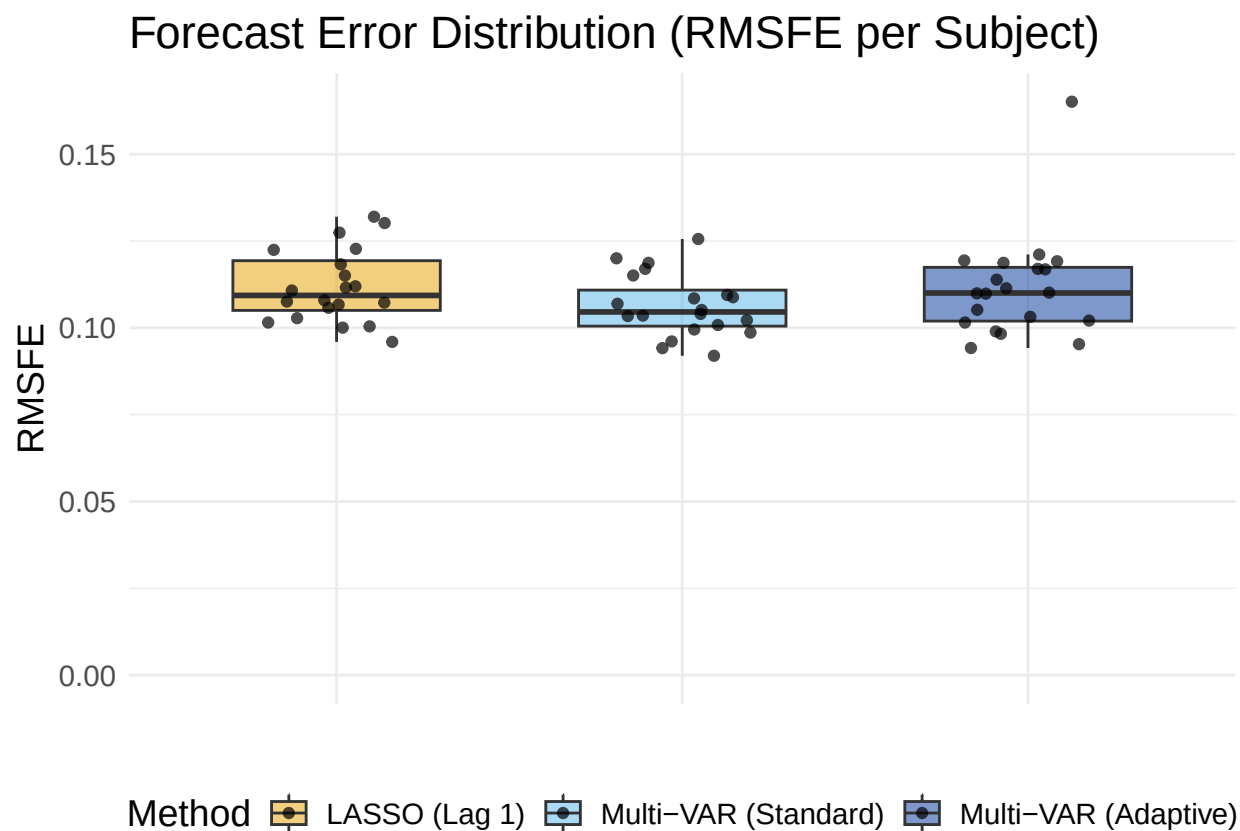
ggplot(rmsfe_dat, aes(x = Method, y = RMSFE, fill = Method)) +
  geom_boxplot(alpha = 0.5, outlier.shape = NA, width = 0.6) +
  geom_jitter(width = 0.2, height = 0, size = 1.5, alpha = 0.7) +
  scale_fill_manual(values = color_lab) +
  expand_limits(y = 0) +

  labs(
    title = "Forecast Error Distribution (RMSFE per Subject)",
    #subtitle = "Lower is Better; Boxplots show variability across subjects",
    y = "RMSFE",
    x = ""
  ) +
  theme_minimal() +
  theme(
    axis.text.x = element_blank(),
    axis.ticks.x = element_blank(),
```

```

legend.position = "bottom",
text = element_text(size = 14)
)

```



```

save.image(file="setting11.RData")

```